

Virtualization and Introduction to Public Cloud

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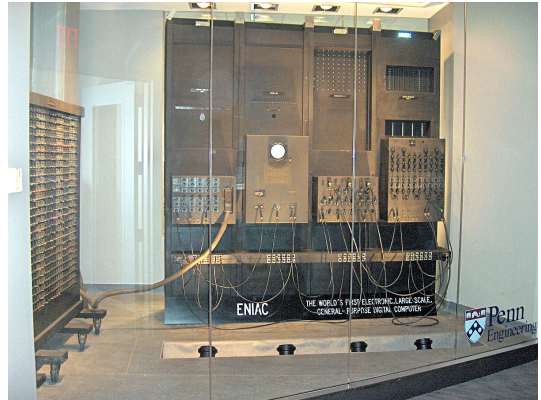
- 1 Introduction
- 2 Virtualization
- 3 Hardware Virtualization mechanisms
- 4 VM Management
- 5 Containers
- 6 Cloud Computing

Outline

- ## 6 Cloud Computing

3rd Industrial revolution: The digital revolution (1945)

- Semiconductors
- Mainframes
- Computers
- Gaming
- Internet



4rd Industrial revolution: The convergence revolution (2023)

- Cloud Computing
- Data analytics
- Smartcities
- Artificial Intelligence



Outline

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- 2 **Virtualization**
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What is virtualization...

Virtualization at a glance

An abstraction mechanism to manage
(by partitioning, by merging,...) physical resources

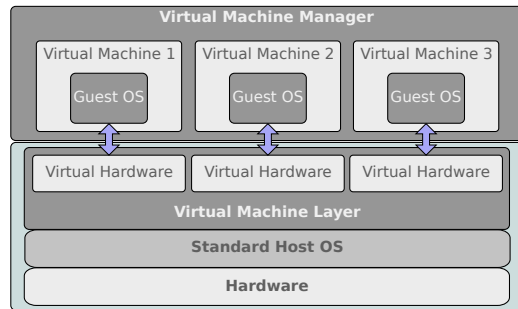
- Virtualization is based on the creation of one (or more) virtual representations of a particular resource
- Examples
 - Local Area Network (VLAN)
 - Web Virtual Hosts
 - Storage Virtualization (e.g., LVM, Cloud storage)
 - Data Virtualization (e.g., seamless access to information)
 - **Hardware Virtualization**

Outline

- 1 Introduction
- 2 Virtualization
- 3 **Hardware Virtualization mechanisms**
 - Full Virtualization
 - Paravirtualization
 - Partial virtualization
- 4 VM Management
- 5 Containers

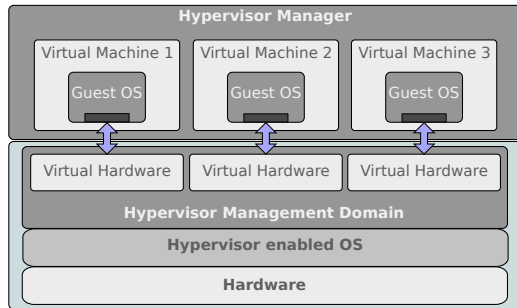
Full Virtualization – Overview

- Complete emulation of hardware components
 - Using the legacy OS as proxy to the hardware
- Transparent for guest OS



Paravirtualization

- Concurrent execution of different OS: Management Domain controls the rest of OS
- Requires support of the hardware and the OS



Paravirtualization – Functionalities and Examples

Functionalities

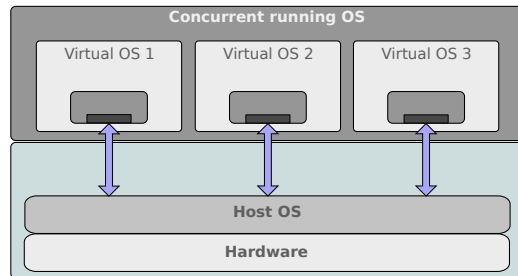
- Memory partitioning
- CPU partitioning
- Network card separation
- Controlled bus access

Examples

- Xen

Partial Virtualization

- Collaboration between host and guests
 - Direct access to the hardware from the guests
 - Can run in userspace
- Requires support of the OS
 - Host and guests use the same OS



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- 3 Hardware Virtualization mechanisms
- 4 **VM Management**
 - **Integrated Management Solutions**
 - **Backups**
- 5 Containers
- 6 Cloud Computing

Integrated Management Solutions

- libvirt: `virsh`, `virt-manager` <http://www.libvirt.org>
- VMWare vSphere <http://www.vmware.com/products/vsphere>
- OpenNebula: <http://www.opennebula.org/>
- OpenStack: <http://www.openstack.org/>

Backup mechanisms

Full Hardware backup

- Use available facilities within the Virtualization Software
- Use a master copy of the system

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What is a container

- A lightweight, standalone package
- It includes everything needed to run an application
 - The application
 - The runtime dependencies
 - *Necessary* system tools
 - Libraries
 - Settings
- There are plenty of public repositories with public images
 - DockerHub

- Mimic a virtual machine
 - Because they run in separate memory, CPU, and Storage spaces
- Are immutable
 - All changes are lost if the container is recreated/updated
- They work in layers



- Containers create local virtual network
 - By default in the range `172.16.0.0/12`
 - But using /16 prefixes → from `.16` to `.31`
- Physical host acts as a virtual router/switch
- Security is automatically enforced through `iptables`
 - By default all incoming traffic is denied unless explicitly stated

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Outline

- 1 Introduction
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- 4 VM Management
- 5 Containers
- 6 Cloud Computing
 - Rationale
 - Delivered Services

Data center and cloud impact

- Present in our day to day
- It affects our economy
- It forces changes in our business models



What is cloud? NIST says:

What is cloud from NIST

cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction

Goals of adopting the cloud

- We want to reduce:
 - **Time-to-market**
 - **Risk**
 - **Costs**: initial investment vs monthly fee
- Concentrate on the business not the infrastructure
- Do not mind about the company technological resources limitations
- Let experts handle the infrastructure

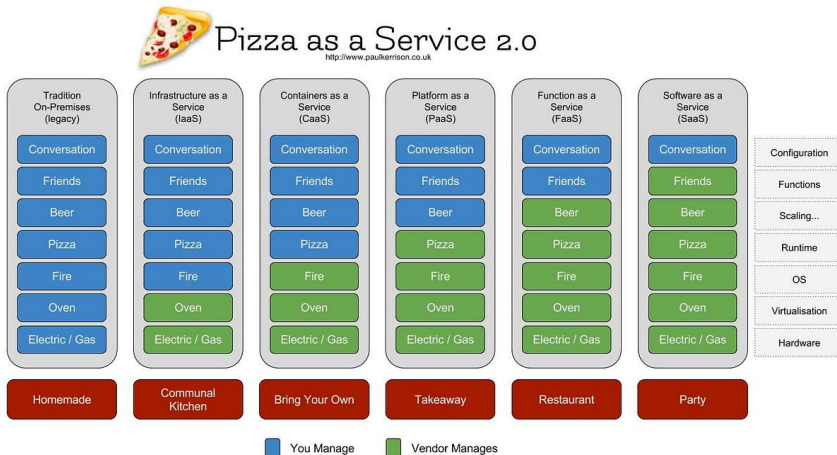
Risks and problems

- What if connectivity drops?
- What if the system fails and all is lost?
- Does regulation allow me to publish this information there?
- How private is my data
- You still need to secure the system
- Lack of control
- Vendor-lock-in

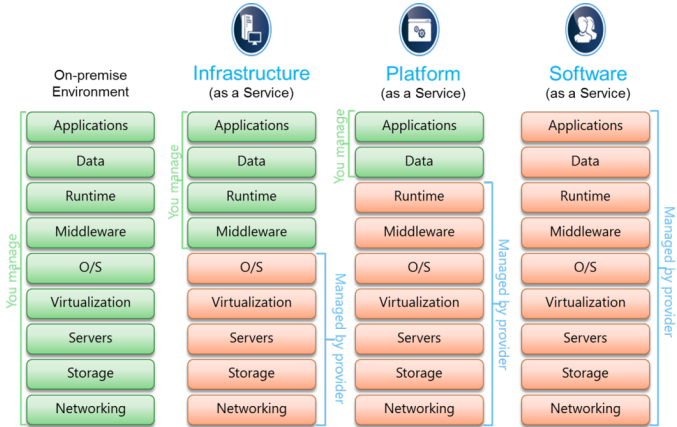
How can we use the cloud?

- We have various models:
 - **Public Cloud**: all infrastructure on the cloud
 - **Private Cloud**: company control of the infrastructure
 - **Community Cloud**: an in-between
- But all deployments may be hybrid, where part of the infrastructure is on a different place

Then the question is: *How much control do we want to keep?*



NIST Service Models



Different providers of Public Cloud



IBM Cloud

Software as a Service (SaaS)

- Software licensed on a subscription based fee model
 - Through periodic fees
 - Through advertisement
- Mostly web based
- Examples
 - GMail
 - Facebook
 - Whatsapp

- Computing platform
- Customer deploys application using service provider features into the provider's premises
- Provider offers storage, computation, memory, networks, and other necessary facilities
- Preconfigured environment for easy testing and development
- Examples
 - Amazon Web Services (AWS)
 - Heroku

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Outline

AWS Basics

- **Regions:** physical locations of the datacenters
- **Availability Zones (AZs):** different datacenters in the same location (physically separated)
- **Virtual Private Clouds (VPCs):** where resources are interconnected

Keys of the cloud

- Highly distributed infrastructure into regions and AZs
- Abstraction of the configuration details through services

AWS Regions



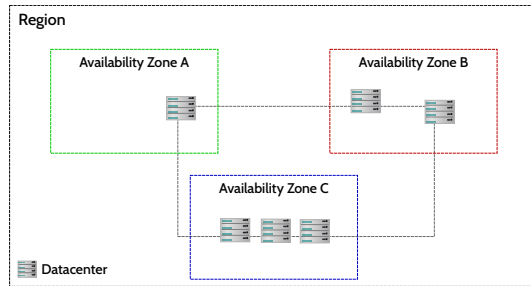
AWS Some numbers²

- 39 Regions
- Using 123 availability zones
- 750+ Cloudfront POPs and 15 Regional edge caches
- 43 Local Zones
- 33 Wavelength Zones for low latency applications

² AWS Global Infrastructure

Overall AWS structure

- Each region is built into *Availability Zones* (AZ) - **Datacenters**
- Depending on the *Region* it may have different amount of AZs
- Connected through a low latency high bandwidth redundant links

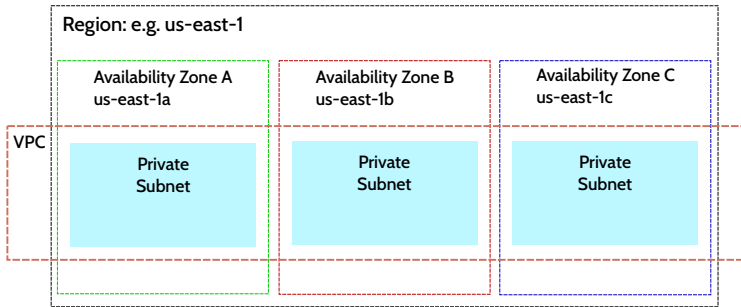


AWS Network characteristics

- All regions are interconnected through the internal Amazon network
- All regions can be directly accessed through the Internet

Overall AWS structure

- A Virtual Private cloud is an abstraction of a subnetwork
- It goes across AZs within the same region
- But a subnet needs to be attached to a single AZ



Most relevant services

- AWS IAM
- AWS EC2
- AWS S3
- Amazon RDS
- Serverless Services



AWS Identity and Access Management (IAM)

IAM Definition

IAM provides fine-grained permissions to AWS services and resources

- Manages access to resources:
 - **Who?** → Users / Groups of users
 - **can access?** → permissions and privileges
 - **What?** → resources within your organization
- It uses *Attribute Based Access Control*
- It manager per account identities and ties its permissions to them across AWS
- It abstracts permissions in the form of *roles*
- Allows the enforcement of "*Least privilege principle*"



AWS Elastic Cloud Computing (EC2)

- Provides versatile computing nodes (instances)
- The admin can decide how much memory, disk, CPU, and how many virtual networks an instance will have
- There is a broad selection of preinstalled operating systems
- Amazon handles the infrastructure
- The admin manages:
 - The operating system
 - The applications
 - The configuration
 - The Security Groups
 - The permissions, ...



EC2

- ## ^aAWS Instance Types

- Assigned to one or more subnets
- Private or public addresses
- Attached to one or more **security groups** → one per network interface

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AWS Security Groups

Security Groups

A security group controls the network traffic that is allowed to reach and leave the resources that it is associated with. When you create a VPC, it comes with a default security group^a

^a[AWS Documentation](#)

- It must be attached to a VPC → it can only control accesses within that VPC
- They are **stateful**: the response traffic will always be allowed
- They can be applied to virtually anywhere within VPC³
- Only users allowed from IAM can modify security groups

³There are notable exceptions to this as stated [here](#)

AWS Simple Storage Service (S3)

Simple Storage Service (S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. You can use Amazon S3 to store and retrieve any amount of data at any time, from anywhere^a

^aMore info [here](#)

AWS Simple Storage Service (S3)

- Object oriented storage system
- Each element (object) is composed of a file and metadata about it
- Each object may have a different set of permissions
- The objects are stored into buckets
 - Each bucket allows to control general permissions
 - Who can create, delete, or list objects in the bucket
- You may regard the bucket as a kind of *directory*



Amazon Relational Database Service (RDS)

- Easy to manage relational database
- Allows to use **various engines: Aurora MySQL, PostgreSQL, Oracle**
- **Automatic failover**: the database is automatically replicated to different AZs
- DB infrastructure management with AWS expertise



amazon
RDS

Managing RDS

The admin just needs to care about the database content and who can access what, the rest is taken care of by AWS staff

What serverless means?

What is a Serverless system?

Serverless computing is a method of providing backend services on an as-used basis. A serverless provider allows users to write and deploy code without the hassle of worrying about the underlying infrastructure. A company that gets backend services from a serverless vendor is charged based on their computation and do not have to reserve and pay for a fixed amount of bandwidth or number of servers, as the service is auto-scaling. Note that despite the name serverless, physical servers are still used but developers do not need to be aware of them^a

^aMore information [here](#)

AWS Lambda

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Just upload a ZIP file with the code or container image, the system will allocate the resources and run the code whenever there is an input or a given event

¹More information in [here](#)

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Why Pair API Gateway with AWS Lambda?

Operational Efficiency

- **Zero Infrastructure Management:** Fully managed by AWS; no OS patching or server maintenance
- **Highly Elastic Scaling:** Automatically scales from zero to thousands of concurrent requests without manual intervention
- **Cost-Effective:** Pay only for API gateway calls and execution time (billed in milliseconds)

Edge Security & Control

- **Throttling:** Apply rate-limiting at the API stage level to prevent DDoS attacks and Lambda starvation
- **Native Authorization:** Authenticate users via Cognito User Pools, IAM roles, or custom **Lambda Authorizers**
- **API Caching:** Cache endpoint responses to reduce execution costs and latency

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Messaging functionalities

- SQS
- SNS
- Step-functions

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AWS example - Architecture

